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Computer Laboratory Report: Finite Element Method

Introduction

This report presents a finite element analysis (FEA) of the deflection of a fishing rod when a fish is hooked. The study aims to determine the deformation of the rod under the applied load, i.e., the force exerted by the fish, by calculating nodal displacements and visualizing both the undeformed and deformed configurations. The rod is modeled as a two-dimensional (2D) beam structure, discretized into multiple beam elements interconnected by nodes. It has a total length of 2 meters, a radius of 0.05 meters, and features a circular cross-section.

The fishing rod consists of four elements with five nodes, and its total length is evenly divided among the elements. When subjected to an external load, i.e., the pull from the fish, the rod undergoes deformation in the horizontal direction (x-axis), the vertical direction (z-axis), and rotation in the y-direction. This indicates that the system's structure is in a 2D frame situated on the xz-plane where each node has three degrees of freedom (DOFs): horizontal displacement (u_x), vertical displacement (w_z), and rotation (θ_y).

The study focuses on constructing and assembling stiffness matrices to model how the fishing rod resists deformation under the applied load. The methodology involves using transformation matrices to convert local stiffness matrices into the global coordinate system while considering appropriate boundary conditions (BCs). Key objectives include formulating stiffness matrices for single- and multiple-element structures, assembling the global stiffness matrix from local components, applying boundary conditions, solving the rod displacement when a fish is hooked, and generating a plot to visualize the deformation.

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Methodology

1. Local Stiffness Matrix

A stiffness matrix describes how a structure resists deformation when subjected to external forces. Since the fishing rod is modeled as a beam structure, each beam element has its own local stiffness matrix. For a single beam element in a 2D frame, the local stiffness matrix is a 6x6 matrix that relates forces and moments to displacements and rotations at its two nodes. The relationship between forces and displacements, the force equilibrium, can be described by Hooke's law:

$$\vec{F} = \boldsymbol{K}\vec{u} \tag{1}$$

, where $\vec{F}$ is the vector of applied forces, $\boldsymbol{K}$ is the stiffness matrix of the structure, and $\vec{u}$ is the vector of nodal displacements.

2. Transformation to global coordinates

In FEA, the assembly of the global stiffness matrix $\boldsymbol{K_e}$ requires all structural elements to be expressed in a common coordinate system. Since the individual beam elements are defined in their respective local coordinate systems $\boldsymbol{k_e}$, their properties must be transformed into the global coordinate system to ensure consistency in the analysis.

$$\boldsymbol{k_e} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \tag{2}$$

, where E is Young's modulus, A is the cross-section area, I is the inertia, and L is the length of the element.

For each element, with nodes i and $i+1$, the deformation of each DOF in the local and global coordinate system are described, respectively, as:

$$\vec{d}_e = \left[u_{x,i}, w_{z,i}, \theta_{y,i}, u_{x,i+1}, w_{z,i+1}, \theta_{y,i+1}\right]^T \tag{3}$$

$$\vec{D}_e = \left[D_{i,1}, D_{i,2}, D_{i,3}, D_{i+1,1}, D_{i+1,2}, D_{i+1,3}\right]^T \tag{4}$$

The following equations express the transformation of deformations from the global coordinate system to the local coordinate system using trigonometric relationships based on the beam element's angle α.

$$\begin{aligned} u_{x,i} &= D_{i,1}\cos(\alpha) + D_{i,2}\cos(\beta) \\ &= \left\{\cos(\beta) = \cos\left(\frac{\pi}{2} - \alpha\right) = \sin(\alpha)\right\} \\ &= D_{i,1}\cos(\alpha) + D_{i,2}\sin(\alpha) \end{aligned} \tag{5}$$

$$w_{z,i} = -D_{i,1}\sin(\alpha) + D_{i,2}\cos(\alpha) \tag{6}$$

$$\theta_{y,i} = D_{i,3} \tag{7}$$

From these equations, the transformation matrix $\boldsymbol{T}$ that linearly converts global coordinates to local coordinates is obtained:

$$\boldsymbol{T} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 & 0 & 0 & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\alpha) & \sin(\alpha) & 0 \\ 0 & 0 & 0 & -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

Thus, the deformation in the global coordinate system is defined by the following linear matrix-vector multiplication:

$$\vec{d}_e = \boldsymbol{T}\vec{D}_e \tag{9}$$

The same operation is applicable to the local and global force vectors:

$$\vec{f}_e = \boldsymbol{T}\vec{F}_e \tag{10}$$

According to (1)—the relationship between force and displacement—the global stiffness matrix for each element $\boldsymbol{K_e}$ is described by:

$$\vec{F}_e = \boldsymbol{K_e}\vec{D}_e \tag{11}$$

This relationship is derived by applying (10), (1), and (9), leading to the global stiffness matrix for each element being obtained through matrix multiplication:

$$\vec{F}_e = \{(10)\} = \boldsymbol{T}^T\vec{f}_e = \{(1)\} = \boldsymbol{T}^T\boldsymbol{k_e}\vec{d}_e = \{(9)\} = \boldsymbol{T}^T\boldsymbol{k_e}\boldsymbol{T}\vec{D}_e \Longrightarrow \boldsymbol{K_e} = \boldsymbol{T}^T\boldsymbol{k_e}\boldsymbol{T}$$

$$\boldsymbol{K_e} = \boldsymbol{T}^T\boldsymbol{k_e}\boldsymbol{T} \tag{12}$$

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3. Assembly of the Global Stiffness Matrix

The global stiffness matrix $\boldsymbol{K_{global}}$ is the assembly of transformed local stiffness matrices $\boldsymbol{K_e}$. At shared nodes, i.e., all nodes except the boundary nodes in the fishing rod, multiple elements contribute to the same DOFs due to their respective local stiffness matrices. Consequently, the stiffness contributions from all connected elements are summed at these shared DOFs in the corresponding entries of $\boldsymbol{K_{global}}$, ensuring equilibrium and continuity in the global system.

The size of $\boldsymbol{K_{global}}$ is determined by the total number of DOFs. Since all five nodes have three DOFs, as defined by (5)-(7), the total number of DOFs is 15. As a result, the matrix's size is 15x15. The square (quadratic) form results from the balance between the number of unknown displacements, i.e., DOFs, and the number of node equilibrium equations, i.e., equations (5)-(7) per node. According to (1), the global stiffness matrix is described by:

$$\vec{F}_{global} = \boldsymbol{K_{global}}\vec{D}_{global} \tag{13}$$

, where $\vec{D}_{global}$ is the displacement vector, containing all DOFs (D_i, $i \in \{1, \dots, 15\}$).

4. Boundary Conditions and System Reduction

As a first boundary condition (BC1), the first node is fixed, i.e., no deformation can occur. Consequently, all its DOFs are zero.

$$u_{x,1} = 0 \tag{14}$$

$$w_{z,1} = 0 \tag{15}$$

$$\theta_{y,1} = 0 \tag{16}$$

Furthermore, the 20 N load applied in the negative z-direction at the free end of the fishing rod defines a second boundary condition (BC2) as follows: (a) the force in the x-direction is zero, (b) the force in the z-direction is -20 N, and (c) the torque, given by the vector product of the force ($F = -20$ N) and the length of the last beam element ($L = 0.5$ m), is -10 Nm in the positive y-direction.

$$F_{5,x} = 0 \tag{17}$$

$$F_{5,z} = -20 \tag{18}$$

$$M_{5,y} = -10 \tag{19}$$

As a result of BC1, i.e., defined by equations (14)-(16), the remaining 12 DOFs—corresponding to the "free" nodes—reduce the global system (12) to a 12x12 system of equations used to solve for the nodal deformations. In this system, the deformation vector $\vec{D}_{free}$ contains the unknown deformations D_i, where $i \in \{1, \dots, 15\}$. Additionally, BC2, i.e. defined by equations (17)-(19), is incorporated into the force vector $\vec{F}_{free}$.

$$\vec{F}_{free} = \boldsymbol{K_{free}}\vec{D}_{free} \quad (20)$$

5. Calculation of deformations

In MATLAB, the reduced system of linear equations (20) was efficiently computed using the backslash operator (\). These nodal deformations were added to the initial (undeformed) node positions to determine the final (deformed) configuration of the fishing rod.

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Result

1. Visualization of Fishing Rod Deflection Under Load

Finally, the undeformed and deformed configurations of the fishing rod were plotted, as shown in Figure 1, providing a clear visualization of the rod's behavior under the applied load in the xz-plane. However, the rotation, i.e., θ_y, is not included in the visualization, as a three-dimensional (3D) representation of the deformed shape is needed to accurately observe the rotational effects.

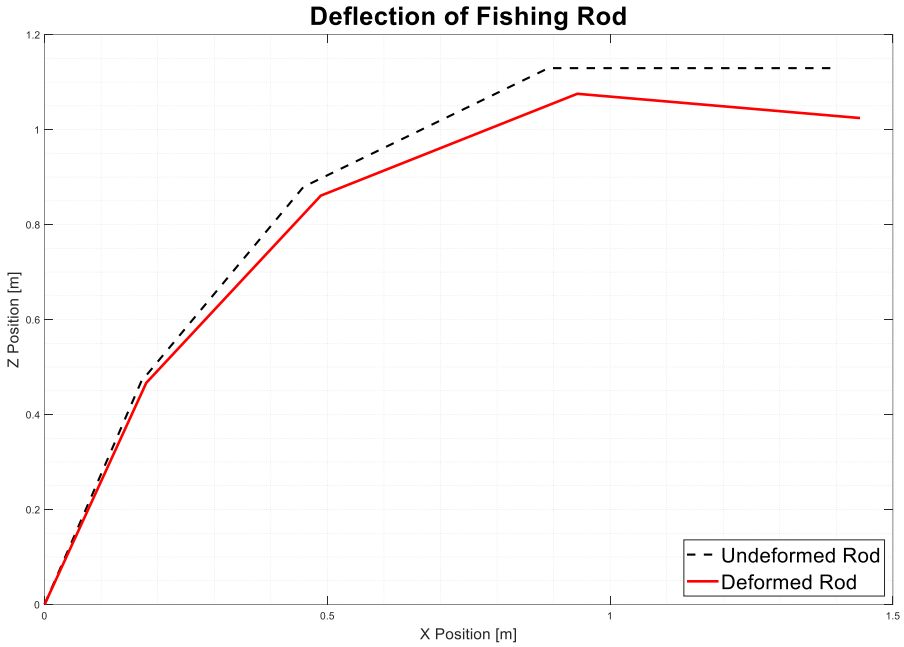


Figure 1. Deflection of Fishing Rod.

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2. Deformation Analysis of a Rod Under Applied Force

In Table 1, the deformations illustrate how the rod bends in response to the applied force. In the *x*-direction, the gradual increase in deformation indicates a steady displacement along the horizontal axis. Conversely, in the *z*-direction, the deformation reveals a more rapid change, suggesting a stronger influence from the force vector. This increase eventually stabilizes due to the stiffness of the rod.

To achieve a more accurate representation of the rod's elastic behavior, a higher discretization of nodes, i.e., dividing the rod into more elements, would better approximate continuous deformation, as the rod model would be converted from discrete to continuous. Initially, a linear elastic response is expected, which is partially reflected in Table 1, as the deformation in both the *x*- and *z*-directions for the first three nodes follows a linear trend. However, as the table further shows, the deformations ultimately exhibit a nonlinear trend. The overall observations indicate that the rod undergoes elastic bending.

In addition to the displacements, Table 1 also presents the variation in the angle and rotation (deformation) in the *y*-direction, which provides further insights into the deformation behavior. The initial angle of 70° suggests a steep inclination at the first node, but as the rod deforms, the angle gradually decreases, eventually reaching -6.24°. The nonlinear progression in these values further confirms that as the deformation increases, the rod undergoes a more complex rotational response, reinforcing the conclusion that it experiences elastic bending.

Table 1. Deformation in x- and z-directions.

X Position [m]	Deformation in x-direction [m]	Z Position [m]	Deformation in z-direction [m]	Angle [°]	Rotation in y-direction [°]
0	0	0	0	70	0
0.17101	0.17978	0.46985	0.46664	52.893	-2.1072
0.4578	0.4884	0.87942	0.86092	26.053	-3.9472
0.89081	0.94192	1.1294	1.0754	-5.3671	-5.3671
1.3908	1.4419	1.1294	1.0243	-6.2425	-6.2425

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MATLAB code

```
% Yolande Ekpe, aneekpe@kth.se
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clc; clear; close all;


% === PARAMETERS ===

num_elements = 4;                               % Number of elements
num_nodes = 5;                                  % Number of nodes

L_tot = 2;                                      % m
L = L_tot / num_elements;                       % m
r = 0.05;                                       % m
I = pi*r^4 / 4;                                 % m^4
E = 100e6;                                      % MPa
A = pi*r^2;                                     % m^2

beta = 70;                                      % degrees
lambda = 55;                                    % degrees
delta = 30;                                     % degrees
angles_deg = [beta, lambda, delta, 0];          % Element angles in degrees
angles_rad = deg2rad(angles_deg);               % Element angles in radians

F = 20;                                         % N


% === LOCAL STIFFNESS MATRIX ===

k_e = [
    E*A/L, 0, 0, -E*A/L, 0, 0;
    0, 12*E*I/L^3, 6*E*I/L^2, 0, -12*E*I/L^3, 6*E*I/L^2;
    0, 6*E*I/L^2, 4*E*I/L, 0, -6*E*I/L^2, 2*E*I/L;
    -E*A/L, 0, 0, E*A/L, 0, 0;
    0, -12*E*I/L^3, -6*E*I/L^2, 0, 12*E*I/L^3, -6*E*I/L^2;
    0, 6*E*I/L^2, 2*E*I/L, 0, -6*E*I/L^2, 4*E*I/L
];


% === TRANSFORMATION MATRICES ===

T_matrices = cell(1, num_elements);
for i = 1:num_elements
    theta = angles_rad(i);
    T = [
        cos(theta), sin(theta), 0, 0, 0, 0;
        -sin(theta), cos(theta), 0, 0, 0, 0;
        0, 0, 1, 0, 0, 0;
        0, 0, 0, cos(theta), sin(theta), 0;
        0, 0, 0, -sin(theta), cos(theta), 0;
        0, 0, 0, 0, 0, 1
    ];
    T_matrices{i} = T;
end
```

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```
% === SETUP REDUCED SYSTEM ===

% In the 2D-frame (x- and z-axis), each node has 3 DOFs: horizontal
% displacement (u_x), vertical displacement (w_z), and rotation (theta_y).
% The third independent variable, the force torque, is always in the
% y-direction, regardless of the force direction. The total degrees of
% freedom (DOFs) in the rod is FORMULA:
DOFs = 3*num_nodes;

% A vector of all DOF indices from 1 to DOFs.
DOFs_vec = 1:DOFs;

% BOUNDARY CONDITION (1):
% The first node is fixed, which means that all of its DOFs are
% constrained, i.e., no displacements (u_x, w_z) and no rotation (theta_y).
DOFs_fixed = DOFs_vec(1:3);

% Removing the fixed DOFs from the complete list of DOFs.
DOFs_free = DOFs_vec(4:end);


% === GLOBAL STIFFNESS MATRIX ===

% Initialization of the global stiffness matrix, which is the sum of all
% the local transformed matrices.
K_global = zeros(DOFs, DOFs);

% Transforming the local matrices to global matrices.
K_transformed = cell(1, num_elements);
for i = 1:num_elements
    T = T_matrices{i};
    K_transformed{i} = T'*k_e*T;
    first_node = (3*(i-1)+1):(3*(i-1)+3);       % DOFs of node i
    second_node = (3*i+1):(3*i+3);              % DOFs of node (i+1)
    index = [first_node, second_node];          % Concatenate a 1x6 vector

    % Selecting the submatrix of the global stiffness matrix corresponding
    % to index, i.e., the DOFs associated with the current element.
    K_global(index, index) = K_global(index, index) + K_transformed{i};
end

% Extracting the submatrix of the global stiffness matrix used to solve
% for the displacements at the free DOFs.
K_free_nodes = K_global(DOFs_free, DOFs_free);


% === FORCE VECTOR ===

% BOUNDARY CONDITION (2):
% The force is applied in negative z-direction.
F_vec = zeros(DOFs, 1);         % Initialization
F_vec(end-2) = 0;               % Force in x-direction
F_vec(end-1) = -F;              % Force in z-direction
F_vec(end) = -F*L;              % Torque

% Extracting the entries of the force vector used to solve for the
% displacements at the free DOFs.
F_free_nodes= F_vec(DOFs_free);
```

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```
% === UNDEFORMED ROD ===

% Defining initial node position vectors.
nodes_x = zeros(1, num_nodes);      % Initialize x-coordinates
nodes_z = zeros(1, num_nodes);      % Initialize z-coordinates

% BOUNDARY CONDITION (1):
% First (fixed) node is at the origin.  u_x = 0, w_z = 0, theta_y = 0
fixed_node_x = nodes_x(1);
fixed_node_z = nodes_z(1);

% Computing initial node positions based on element length and angle.
for i = 2:num_nodes
    nodes_x(i) = nodes_x(i-1) + L * cos(angles_rad(i-1));
    nodes_z(i) = nodes_z(i-1) + L * sin(angles_rad(i-1));
end


% === DEFORMED ROD ===

% Computing the vector of global DOF indices for each node: horizontal
% displacement (u_x), vertical displacement (w_z), and rotation (theta_y).
D_vec = zeros(DOFs, 1);
D_vec(DOFs_free) = K_free_nodes\F_free_nodes;

% Computing the deformed rod based on node displacement.
deformation_x = D_vec(1:3:end);
deformation_z = D_vec(2:3:end);
deformation_y = D_vec(3:3:end);
displacement_x = nodes_x + deformation_x';
displacement_z = nodes_z + deformation_z';


% === DEFORMATION TABLE ===

deformation_tab = table( ...
    displacement_x', displacement_z', ...
    'VariableNames', ...
    {'Deformation in x-direction', ...
    'Deformation in z-direction'});
disp(deformation_tab);


% === PLOT DEFORMED ROD ===

figure(1);
plot(nodes_x, nodes_z, 'k--', 'LineWidth', 2); hold on;
plot(displacement_x, displacement_z, 'r-', 'LineWidth', 2.5);
xlabel('X Position [m]', 'FontSize', 15);
ylabel('Z Position [m]', 'FontSize', 15);
title('Deflection of Fishing Rod', 'FontSize', 25);
legend('Undeformed Rod', 'Deformed Rod', ...
    'Location', 'southeast', ...
    'Fontsize', 20);
grid minor;
hold off;
```